MAGNATE ACADEMY

TECHNICAL ROUND 3 – DATA SCIENCE

Candidate Name:
Date:

Total Marks: 50

Instructions

- Answer all questions.
- Write Python code wherever requested.
- Give clear explanations and examples where appropriate.
- The practical task will be evaluated separately as a complete Data Science workflow.

PART 1 – PYTHON | 15 MARKS

A. Python Coding | 10 Marks

1. Count the Frequency of Each Character – 2 Marks

Write a Python program to count the frequency of each character in a given string.

Example:

Input: `"programming"`

Output: `{'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}`

2. Second Largest Number – 2 Marks

Write a Python program to find the second-largest distinct number in a given list.

Example:

Input: `[10, 5, 8, 20, 20, 15]`

Output: `15`

Condition: Do not use the built-in `sort()` or `sorted()` functions.

3. Remove Duplicate Elements – 2 Marks

Write a Python program to remove duplicate elements from a list without changing their original order.

Example:

Input: `[1, 2, 3, 2, 4, 1, 5]`

Output: `[1, 2, 3, 4, 5]`

Condition: Do not use the built-in `set()` function.

4. Two Sum – 2 Marks

Given an array of integers `nums` and an integer `target`, return the indices of the two numbers such that they add up to the target.

You may assume that each input has exactly one solution, and you may not use the same element twice.

Example:

Input: `nums = [2, 7, 11, 15], target = 9`

Output: `[0, 1]`

Explanation: `nums[0] + nums[1] = 2 + 7 = 9`.

5. Valid Palindrome – 2 Marks

Given a string `s`, determine whether it is a palindrome. A palindrome reads the same forwards and backwards after converting all uppercase letters to lowercase and removing all non-alphanumeric characters.

Return `True` if the string is a palindrome; otherwise, return `False`.

Example 1:

Input: `s = "A man, a plan, a canal: Panama"`

Output: `True`

Example 2:

Input: `s = "race a car"`

Output: `False`

B. Python Theory | 5 Marks

1. What is the difference between a list and a tuple in Python? Give a suitable example of each.
2. What is the difference between mutable and immutable data types in Python? Give two examples of each.
3. What is exception handling in Python? Explain the purpose of `try`, `except` and `finally` with a suitable example.
4. What is list comprehension in Python?
5. What is a lambda function in Python? Explain how it differs from a normal function with a suitable example.

PART 2 – MACHINE LEARNING | 5 MARKS

1. What is a Confusion Matrix? Explain True Positive, True Negative, False Positive and False Negative.
2. What is Gradient Descent in Machine Learning? How does it help a model learn?
3. What is the difference between Classification and Regression in Machine Learning? Give one example of each.
4. What is Feature Engineering in Machine Learning? Why is it important, and give a suitable example.
5. What is Regularisation in Machine Learning? Explain the difference between L1 and L2 Regularisation.

PART 3 – MACHINE LEARNING PRACTICAL | 15 MARKS

Complete Data Science Project

Task

Download a **new and suitable dataset** and build a complete Machine Learning project.

Your project must include:

Requirement	Marks

Dataset selection and loading	2
Data preprocessing	3
Exploratory Data Analysis and visualisation	2
Feature/target selection and train-test split	2
Model selection and training	2
Model evaluation and interpretation	2
GitHub repository and project documentation	2
Total	**15**

GitHub Requirement

Create a GitHub repository named : `Technical round 1`

Push the complete project to the repository, including the **source code, preprocessing, visualisations, model training, evaluation and** `README.md`.

Note: All stages must be completed in the given order. If any stage is skipped, the marks for that stage and all subsequent stages will be zero, even if the subsequent stages are completed. For example, skipping Stage 3 results in 0 marks for Stages 3–7.

PART 4 – DEEP LEARNING | 5 Marks

1. What is the Vanishing Gradient Problem in Deep Learning? How can it be addressed?
2. What is the difference between Sigmoid, Tanh, and ReLU activation functions?
3. What is the difference between an Epoch, a Batch, and an Iteration during neural network training?
4. What is an Optimiser in Deep Learning? Explain the role of SGD and Adam.
5. What are the default values of `epochs` and `batch_size` when using Keras `model.fit()` if

PART 5 – NLP | 5 Marks

1. What is the difference between Stemming and Lemmatization in NLP? Give a suitable example of each.
2. What is the difference between Bag of Words (BoW) and TF-IDF in NLP?

3. What are Stop Words in NLP? Why are they sometimes removed during text preprocessing?
4. What are Word Embeddings in NLP? Explain how Word2Vec represents words numerically.
5. What is the difference between Unigrams, Bigrams and Trigrams in NLP? Give a suitable example of each.

PART 6 – Gen Ai | 5 Marks

1. What is Generative AI? How is it different from Traditional AI?
2. What is a Large Language Model (LLM)? Give some examples of LLMs.
3. What is Retrieval-Augmented Generation (RAG)? Why is it used in Generative AI applications?
4. What is Metadata in Generative AI? Why is it important when storing and retrieving documents in a RAG system?
5. What is Source Tracking and Citation in Generative AI?